

a) [0,6] b) [0,2,3,6] c) [2,4,6,8] d) [3,6,9]

Q116. What does `list(map(str, [1,2,3]))` return?

a) [1,2,3] b) ["1","2","3"] c) "123" d) ["str(1)","str(2)","str(3)"]

Q117. What does `list(filter(None, [0,1,"","hello",None,3.14]))` return?

a) [0,"hello",3.14] b) [1,"hello",3.14] c) [0,1,"",None] d) [None]

Q118. What is `sum(i for i in range(10) if i%2==0)`?

a) 20 b) 25 c) 30 d) 0

Q119. What does the for-loop on empty list print?

`for i in []: print(i)`

`else: print("empty")`

a) Nothing b) "empty" c) Error d) 0

Q120. What does `match = "low" if x<5 else "mid" if x<10 else "high"` give for `x=3`?

a) "low" b) "mid" c) "high" d) Error

Q121. Which correctly iterates over a dict's key-value pairs?

a) `for key,value in d:` b) `for key in d.values()` c) `for key,value in d.items()` d) `for i in range(len(d))`

Q122. What does `[x**2 for x in range(10)][-1]` return?

a) 81 b) 100 c) 9 d) Error

Q123. Time complexity of searching a sorted vs unsorted list?

a) $O(\log n)$ vs $O(n)$ with binary search b) $O(n)$ vs $O(n)$ always for lists c) $O(1)$ vs $O(n)$ d) $O(\log n)$ vs $O(\log n)$

Q124. What does `any(x>5 for x in [1,3,7,2])` return?

a) True b) False c) 7 d) [7]

Q125. How many times does `(i==j)` trigger in two nested `range(5)` loops?

a) 25 b) 5 c) 10 d) 0

TOPIC 6: FUNCTIONS, LAMBDA & MODULES (Q126–Q150)

Q126. What does a function return if it has no return statement?

a) 0 b) "" c) None d) Error

Q127. What does `greet("Ahmad")` print if `def greet(name, msg="Hello"): return f'{msg}, {name}!'`?

a) ", Ahmad!" b) "Hello, Ahmad!" c) Error d) "greet, Ahmad!"

Q128. What does *args capture in a function signature?

- a) Keyword arguments as a dict b) Positional arguments as a tuple c) Positional arguments as a list
- d) All arguments together

Q129. What does f(1,2,a=3,b=4) return for def f(*args,**kwargs): return len(args),len(kwargs)?

- a) (2,2) b) (4,0) c) Error d) (2,4)

Q130. What is the key difference between print(x) and return x in a function?

- a) No difference b) print() displays to stdout; return passes value to caller c) return() displays to stdout
- d) print() stores the value; return() discards it

Q131. What is the output of calling count_up(1) then count_up(2) with def count_up(n, acc=[]): acc.append(n); return acc?

- a) [1] then [2] b) [1] then [1,2] c) Error d) [1] then [2,1]

Q132. What is the LEGB scope resolution order?

- a) Global→Local→Built-in→Enclosing b) Local→Enclosing→Global→Built-in c) Built-in→Global→Enclosing→Local
- d) Local→Global→Enclosing→Built-in

Q133. What does lambda x,y: x if x>y else y do?

- a) Returns the smaller of x,y b) Returns the larger of x,y c) Returns True if x>y d) Error - lambda can't use if/else

Q134. What does sorted(["banana","apple","cherry"], key=len) return?

- a) ["apple","banana","cherry"] b) ["cherry","banana","apple"] c) ["apple","cherry","banana"] d) ["banana","apple","cherry"]

Q135. What does list(map(lambda x: x*2, range(5))) return?

- a) [0,2,4,6,8] b) [1,2,3,4,5] c) [2,4,6,8,10] d) [0,1,2,3,4]

Q136. What does the closure print?

```
def outer(): x=10; inner=lambda: x; return inner
f=outer(); print(f())
```

- a) Error b) None c) 10 d) NameError

Q137. What does list(filter(lambda x: x%2==0, range(10))) return?

- a) [1,3,5,7,9] b) [0,2,4,6,8] c) [2,4,6,8] d) [0,2,4,6,8,10]

Q138. Where is a function's docstring stored?

- a) In __name__ b) In __doc__ c) In __info__ d) In __help__

Q139. What does math.ceil(3.1) return?

- a) 3 b) 3.1 c) 4 d) Error

Q140. What makes `random.seed(42)` useful in data analytics?

- a) Makes code faster
- b) Makes random output deterministic/reproducible
- c) Limits random values to 42
- d) Resets the module

Q141. What does `from datetime import datetime; datetime.now().year` return?

- a) Error
- b) The current year as an integer
- c) The full datetime string
- d) None

Q142. What is `collections.Counter("hello")`?

- a) 5
- b) `{"h":1,"e":1,"l":2,"o":1}`
- c) `["h","e","l","l","o"]`
- d) Error

Q143. What does `assert 2+2==5, "Math is broken"` do?

- a) Prints "Math is broken"
- b) Raises `AssertionError` with message
- c) Returns False
- d) Does nothing

Q144. What happens when you call `add("x","y")` given `def add(a:int, b:int)->int: return a+b`?

- a) `TypeError` due to type mismatch
- b) "xy" is returned
- c) `AssertionError`
- d) None returned

Q145. What does `collections.defaultdict(list)` provide over a regular dict?

- a) Raises `KeyError` for missing keys
- b) Missing keys automatically get an empty list value
- c) A dict sorted by keys
- d) A read-only dict

Q146. What does `factorial(5)` return for `def factorial(n): return 1 if n<=1 else n*factorial(n-1)`?

- a) 24
- b) 120
- c) 5
- d) infinite recursion

Q147. Which except clause catches `ZeroDivisionError`?

- a) Only ``except ZeroDivisionError:``
- b) ``except Exception:``
- c) Bare ``except:``
- d) All of the above

Q148. What does `math.log(math.e)` return?

- a) 1.0
- b) 0.0
- c) `math.e`
- d) Error

Q149. What does `random.choice([1,2,3,4,5])` do?

- a) Always returns first element
- b) Returns a random element from the list
- c) Shuffles the list then returns None
- d) Returns the median

Q150. What does `reduce(lambda acc,x: acc+x, [1,2,3,4,5])` return?

- a) `[1,2,3,4,5]`
- b) 15
- c) `(1,2,3,4,5)`
- d) Error

119	B	When the iterable is empty, no break occurs, so the else clause still executes.
120	A	$x=3 < 5$ is True so returns "low" immediately. Nested ternary stops at the first True condition.
121	C	.items() yields (key,value) tuples. Iterating d directly yields only keys.
122	A	range(10) ends at 9. $9**2 = 81$. [-1] accesses the last element.
123	A	bisect.bisect_left() on sorted list = $O(\log n)$. Linear scan on unsorted = $O(n)$.
124	A	any() short-circuits: when it finds $7>5=True$, it returns True immediately.
125	B	$i=j$ at (0,0),(1,1),(2,2),(3,3),(4,4) - exactly 5 times. The diagonal of a 5×5 grid.
126	C	Python implicitly returns None when no return statement is reached.
127	B	msg has a default value of "Hello". When not provided, the default is used.
128	B	*args packs extra positional arguments into a TUPLE (not list). **kwargs packs keyword args into a dict.
129	A	1,2 go into args=(1,2): len=2. a=3,b=4 go into kwargs={"a":3,"b":4}: len=2.
130	B	print() is a side effect. return passes the value back so callers can use it. print returns None.
131	B	CRITICAL: mutable default arguments persist across calls. The same list is reused each call.
132	B	LEGB: Local, Enclosing, Global, Built-in. Python searches in this order for

		every name.
133	B	Lambda supports conditional expressions. This lambda returns the maximum of x and y.
134	A	Lengths: apple=5, banana=6, cherry=6. Sorted by length: apple(5) first, then banana,cherry (both 6, stable order).
135	A	range(5)=[0,1,2,3,4]. Each element *2: [0,2,4,6,8].
136	C	Closures capture variables from the enclosing scope. inner captures x=10 from outer's local scope.
137	B	filter keeps elements where the function returns True. Even numbers in range(10): 0,2,4,6,8.
138	B	help(func) reads from func.__doc__. Always write docstrings - they are critical for maintainability.
139	C	ceil() rounds UP to the nearest integer. floor() rounds down. round() rounds to nearest.
140	B	Seeding makes random sequences reproducible - critical for debugging and sharing experiments.
141	B	.now() returns a datetime object. .year extracts the integer year.
142	B	Counter counts element frequencies. "hello" has h×1, e×1, l×2, o×1.
143	B	assert raises AssertionError when the condition is False. The message is included in the error.
144	B	Python type hints are NOT enforced at runtime. The function runs and returns "xy" (string concat).
145	B	defaultdict(list) auto-creates [] for new keys. Ideal for grouping: d[key].append(val) without pre-

checking.

146	B	$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$. Base case $n \leq 1$ returns 1 stopping the recursion.
147	D	All three catch it. Best practice: use <code>'except ZeroDivisionError:'</code> for specific handling; avoid bare except.
148	A	<code>math.log()</code> computes natural logarithm. $\ln(e) = 1$ by definition.
149	B	<code>random.choice()</code> picks ONE element at random. <code>random.choices()</code> allows repetition. <code>random.sample()</code> picks without replacement.
150	B	<code>reduce()</code> applies the function cumulatively: $1+2=3$, $3+3=6$, $6+4=10$, $10+5=15$.